

Why two scheduling lanes are needed

(a) one shared strict-priority lane — fails

higher priority ↑

response blockers (keep draining)

held response

held control command

the response blockers starve the command below, so its release drifts to the response time instead of its own.

(b) two independent lanes — fix

lane 1

resp. blockers

held response

each lane has its own scheduler, so the command releases on its own deadline, independent of the response.

lane 2

cmd blockers

held command

Both lanes are internal paths in the same switch and the same program, not extra devices or a second pipeline.